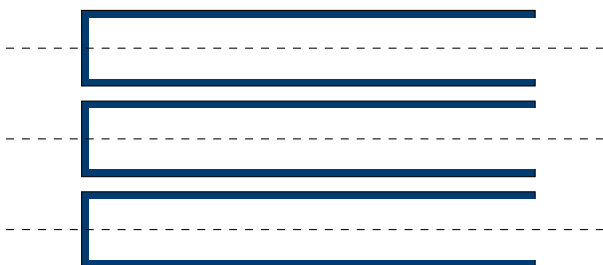


Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

PRELAB EXERCISES (2 points)*This prelab must be completed and handed in to the lab TA at the start of the lab.***Question 1****1 point**

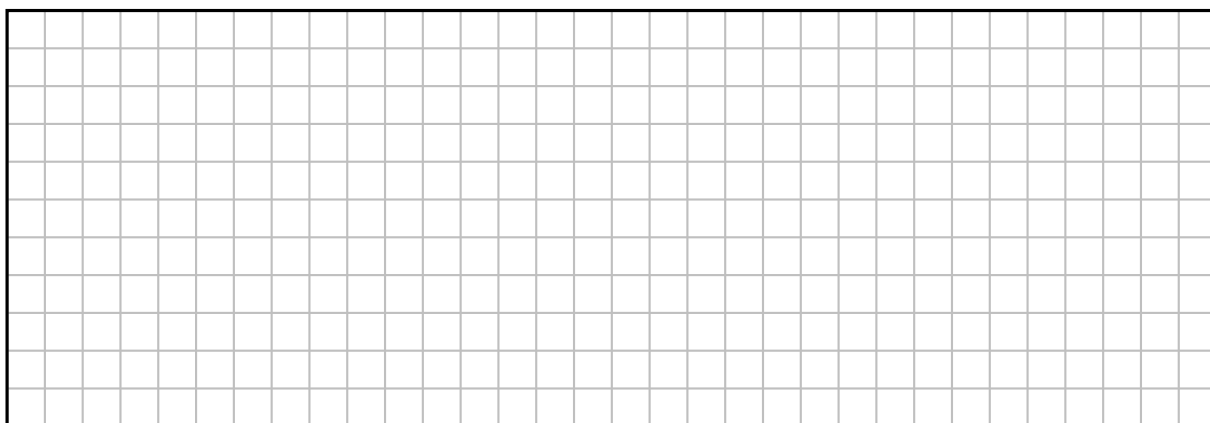
Sound within a hollow tube can form a standing wave. If the end of the tube is open, the standing sound wave must exhibit an antinode there, and if it is closed, the wave must exhibit a node there. Consider a tube of length L with one open and one closed end. Draw the first three harmonics of either the standing *pressure* wave or *displacement* wave of the air in the tube (indicate which one you have chosen to represent). Express each harmonic's wavelength in terms of L .

**Question 2****1 point**

Consider a traveling wave with wavelength $\lambda = 3$ m and frequency $f = 4$ Hz, described by the function

$$f(x, t) = 2 \sin \frac{2\pi}{\lambda} (x + vt).$$

Plot $f(x, 0)$ on the interval $3 \text{ m} \leq x \leq 6 \text{ m}$ below. Then plot the wave at $t = 250$ ms and $t = 500$ ms. Be sure to add labeled axes, and pay attention to units.



What is the period of this wave? Which direction is it moving? How would you change $f(x, t)$ to make the wave move in the opposite direction?